Autodesk®
Scaleform®

Scaleform CLIK AS3 User Guide

CLIK

.

Autodesk®
GAMEWARE

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Autodesk® Scaleform® 4.3

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1 소개

Common Lightweight Interface Kit (CLIK™)

. Getting Started with CLIK Getting Started with CLIK Buttons . 2 CLIK . CLIK CLIK . , .

1.1

CLIK 2.0(AS2) , , Scaleform 4.0 , PC UI . (CPU) , , . UI CLIK UI . , 15 (, , ,) , .

1.1.1 스케일폼 CLIK 에 포함된 것

컴포넌트: UI

Button	Slider
ButtonBar	StatusIndicator
CheckBox	TileList
RadioButton	Label
TextInput	ScrollingList
TextArea	DropdownMenu
ScrollIndicator	NumericStepper
ScrollBar	

Classes: Core system APIs

InputManager	UIComponent
FocusManager	Tween
DragManager	DataProvider
PopUpManager	IDataProvider
IUIComponent	IList
Constraints	IListItemRenderer

Documentation and Example Files:

Getting Started with CLIK
Getting Started with CLIK Buttons
CLIK AS3 API Reference
CLIK AS3 User Guide
CLIK Flash Samples
CLIK Video Tutorials

1.2 UI

UI (Microsoft Visio®)
(visual diagram editor) .

. CLIK
.

UI . (Visio)
. , MovieClip .
, .
.
. ,
. UI
(workflow) .

,

. , ,

( ) .

. ,

.

.

, , UI

. ,

. UI UI (layout)

.

UI

(

) .

.

, . ,

,

.

1.3

, .

. ,

gSkinner.com CLIK

.

Grant Skinner	gskinner.com	CS4 Flash team .
gskinner.com	http://gskinner.com/blog	.

CLIK Flash Studio CLIK

. Scaleform SDK\Resources\AS3\CLIK\components\CLIK_Components_AS3.fla

1.3.1 점검가능 속성

IDE (component inspector panel)

. CS4 Window

Component inspector panel . [Shift+F7]

. (inspectable properties)

.

SWF .

IDE CLIK

. ,

.

1.4

1.4.1 이벤트

, focus .

CLIK .

. .

- x *type*: .
- x *target*: .
- x *currentTarget*: Event .
- x *eventPhase*: . (EventPhase.CAPTURING_PHASE, EventPhase.AT_TARGET, EventPhase.BUBBLING_PHASE).
- x *bubbles*: .
- x *cancelable*: .

ActionScript 3 Adobe ActionScript 3

.

http://help.adobe.com/en_US/FlashPlatform/reference/actionscript/3/flash/events/Event.html

" "

.

Scaleform Scaleform .

Extensions.enabled true Scaleform MouseEvent

MouseEvent MouseEventEx . MouseEventEx

/ .

MouseEventEx

.

1.4.2 Focus

FocusFlash focus . focus

InteractiveObject .

, .

. " " .

CLIK

. TAB Shift+TAB ( )

.

. CLIK CLIK .

CLIK CLIK ,

TAB Shift+TAB

.

Button . Button

,

.

.

, Enter Xbox 360 PS3 A X .

CLIK UI .

그림 1: *Mass Effect* 의 Main Menu Button 예제

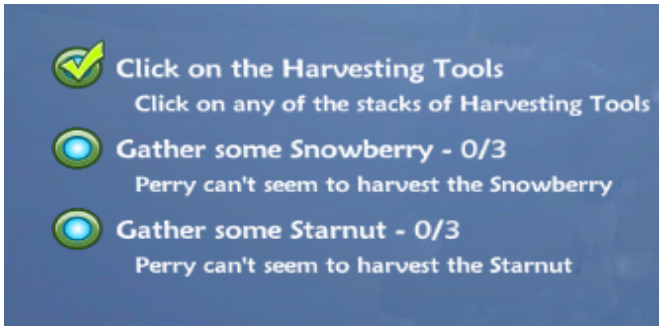


그림 2: *Free Realms* 의 CheckBox 예제

그림 3: *Free Realms* 의 Label 예제

그림 4: *Crysis Warhead* 의 TextInput 예제

5: *Mercenaries 2* TextArea

2.1.1 Button

그림 6: 스킨 없는 버튼

CLIK Button
. Button (scaleform.clik.controls.Button)
textField , (stats) . Button
, ScrollBar Slider thumb .
Button Button .

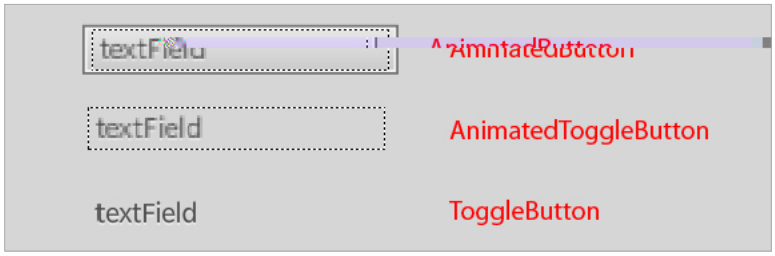


그림 7: AnimatedButton, AnimatedToggleButton, ToggleButton

CLIK Button , ,
. , .
Button.fla ToggleButton, AnimatedButton, AnimatedToggleButton
.

2.1.1.1

Button .
.

2.1.1.2

CLIK Button MovieClip .

- x *textField*: (optional) TextField .
- x *focusIndicator*: (optional) MovieClip . MovieClip .

MovieClip show hide .

over Button . ,

(mouse over)

. focusIndicator .

Button .

.

2.1.1.3 상태

CLIK . .

- x *up*
- x *over* , .
- x *down*
- x *disabled*

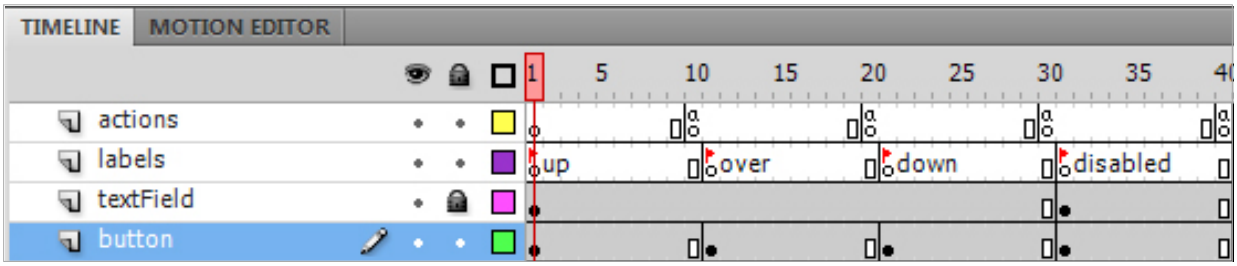


그림 8: Button 타임라인

. , Button

.

(transition) . Getting Started with CLIK Buttons .

2.1.1.4 점검가능 속성

label	.
selected	. . true , false ActionScript .
Toggle	. True .
Visible	false .

2.1.1.5

Event or Event subclass ,

. .

- x type : The type of event.
- x target : The event target.
- x currentTarget : The object that is actively processing the Event object with an event listener.
- x eventPhase: The current phase in the event flow. (EventPhase.CAPTURING_PHASE, EventPhase.AT_TARGET, EventPhase.BUBBLING_PHASE).
- x bubbles: Indicates whether an event is a bubbling event.
- x cancelable : Indicates whether the behavior associated with the event can be prevented.

Button .

.

ComponentEvent.SHOW	(visible property)	true .
ComponentEvent.HIDE	(visible property)	false .
ComponentEvent.STATE_CHANGE	.	
FocusHandlerEvent.FOCUS_IN	focus .	
FocusHandlerEvent.FOCUS_OUT	focus .	
Event.SELECT	.	
MouseEvent.ROLL_OVER	.	

	mouseIdx: ( ). uint . Scaleform MouseEventEx . *buttonIdx*: (zero base index). Scaleform MouseEventEx .
MouseEvent.ROLL_OUT	. *mouseIdx*: ( ). uint . Scaleform MouseEventEx . *buttonIdx*: (zero base index). Scaleform MouseEventEx .
ButtonEvent.PRESS	. *controllerIdx*: ( ). uint . *isKeyboard:* / true false . *isRepeat*: autoRepeating true false .
MouseEvent.DOUBLE_CLICK	. *doubleClickEnabled* true . *mouseIdx*: ( ).

	uint . Scaleform MouseEventEx . *buttonIdx*: (zero base index). Scaleform MouseEventEx .
ButtonEvent.CLICK	. *controllerIdx*: ( ). uint . *isKeyboard:* / true false . *isRepeat*: autoRepeating true false .
ButtonEvent.DRAG_OVER	( ). *controllerIdx*: ( ). uint .
ButtonEvent.DRAG_OUT	( ). *controllerIdx*: ( ). uint .
ButtonEvent.RELEASE_OUTSIDE	, . *controllerIdx*: ( ).

2.1.2.2

CLIK CheckBox MovieClip .

- *textField*: (optional) TextField .
- *focusIndicator*: (optional) MovieClip . MovieClip .
  MovieClip show hide .

2.1.2.3

CheckBox selected .

- *up*
- *over* ,
- *down*
- *disabled*
- *selected_up*
- *selected_over* , .
- *selected_ down*
- *selected_disabled*

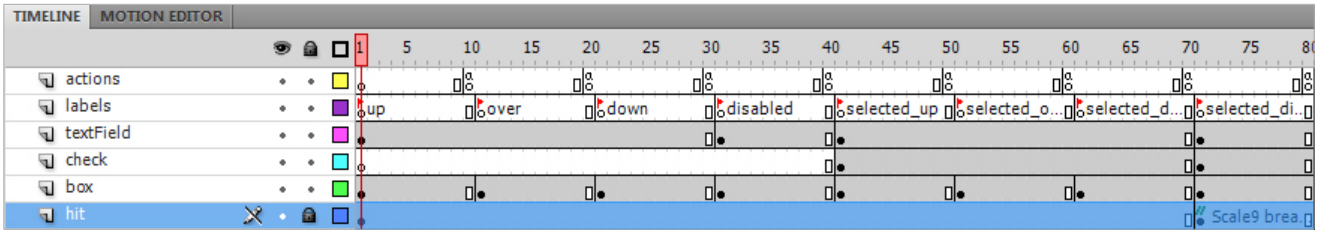


Figure 11: CheckBox timeline.

CheckBox . Button (
CheckBox) Getting Started with CLIK
Buttons .

2.1.2.4

Button CheckBox Button . ,
toggle autoRepeat .

autoSize	, . autoSize autoSize=TextFieldAutoSize.NONE .
Data	. ButtonGroup .
Enabled	false . .
focusable	focus / . focus false tab , focus .
Label	
selected	.

	. true , false ActionScript .
Visible	false

2.1.2.5

. .

- x *type*: .
- x *target*: .
- x *currentTarget*: Event .
- x *eventPhase*: . (EventPhase.CAPTURING_PHASE, EventPhase.AT_TARGET, EventPhase.BUBBLING_PHASE).
- x *bubbles*: .
- x *cancelable*:
- x .

CheckBox .

.

ComponentEvent.SHOW	(visible property) true .
ComponentEvent.HIDE	(visible property) false .
FocusHandlerEvent.FOCUS_IN	focus .
FocusHandlerEvent.FOCUS_OUT	focus .
Event.SELECT	.

ComponentEvent.STATE_CHANGE	.
MouseEvent.ROLL_OVER	. *mouseIdx*: ( ). uint . Scaleform *buttonIdx*: (zero base index). Scaleform MouseEventEx .
MouseEvent.ROLL_OUT	. *mouseIdx*: ( ). uint . Scaleform *buttonIdx*: (zero base index). Scaleform MouseEventEx .
ButtonEvent.PRESS	. *controllerIdx*: ( ). uint . *isKeyboard:* / true false . *isRepeat*: autoRepeating true false .

MouseEvent.DOUBLE_CLICK	. *doubleClickEnabled* true . *mouseIdx*: ( ). uint . Scaleform *buttonIdx*: (zero base index). Scaleform MouseEventEx .
ButtonEvent.CLICK	. *controllerIdx*: ( ). uint . *isKeyboard:* / true false . *isRepeat*: autoRepeating true false .
ButtonEvent.DRAG_OVER	( ). *controllerIdx*: ( ). uint .
ButtonEvent.DRAG_OUT	( ). *controllerIdx*: ( ). uint .
ButtonEvent.RELEASE_OUTSIDE	,

. .

0 3

controllerIdx.

(

). uint .

CheckBox .

```
myCheckBox.addEventListener( Event.SELECT, onCheckBoxToggle);
function onCheckBoxToggle(e:Event) :void {
    // Do something
}
```

2.1.3 Label

Figure 12: Unskinned Label.

CLIK Label (scaleform.clik.controls.Label) MovieClip
(wrapping) textField .
Label textField
. Label textField
. , Label
textField .

Label MovieClip textField .
textField . , Label
AS2 textField diabled
.

Label (constraints) Label (resizing) .
textField Label textField .
Label textField .

2.1.3.1

Label .

2.1.3.2

CLIK Label MovieClip .

- x ***textField***: TextField .

2.1.3.3

CLIK Label disabled 2 .

- x ***default***
- x ***disabled***

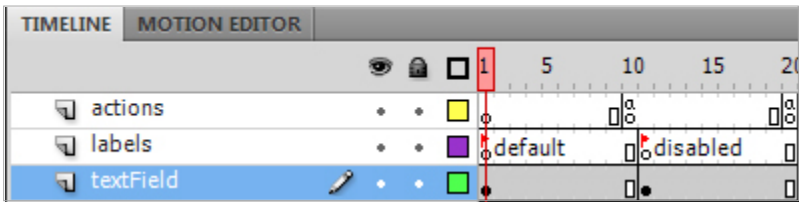


Figure 13: Label timeline.

2.1.3.4

Label .

Text	
Visible	false .
enabled	false .
autoSize	, . autoSize autoSize=TextFieldAutoSize.NONE .

2.1.3.5

. .

x *type*: .

- x *target*: .
- x *currentTarget*: Event .
- x *eventPhase*: . (EventPhase.CAPTURING_PHASE, EventPhase.AT_TARGET, EventPhase.BUBBLING_PHASE).
- x *bubbles*: .
- x *cancelable*:
- x .

Label .

.

ComponentEvent.SHOW	true .
ComponentEvent.HIDE	false .
ComponentEvent.STATE_CHANGE	.

2.1.4 TextInput

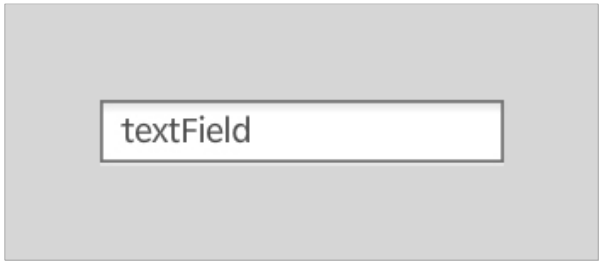


Figure 14: Unskinned TextInput.

TextInput (scaleform.clik.controls.TextInput) textField . Label textField (wrapper) , HTML textField . TextInput textField .

TextInput Label input . Label TextInput textField

. , UI , PC , TextInput
textField .

TextInput textField disable .
TextInput textField
(indicator) . textField AS2
TextInput . TextInput
Flash Studio
. TextInput
.

TextInput HTML (multi paragraph) textField
, , .
(Shift+), (Shift+Delete), (Shift+Copy) (Shift+Insert)
.

TextInput roll over roll out .
actAsButton
. "over" "out" ,
rollOver rollOut . actAsButton "over" "out"
TextInput "default" .
CLIK TextInput
. .

default Text .
defaultText . Default Text ( )
(0xAAAAAA) . TextFormat
defaultTextFormat .

2.1.4.1

TextInput textField .
.

. textField

.

.

2.1.4.2

CLIK TextInput MovieClip .

- x ***textField***: textField .

2.1.4.3

CLIK TextInput disabled 3 .

- x ***default*** enabled
- x ***focused*** , textField
- x ***disabled***

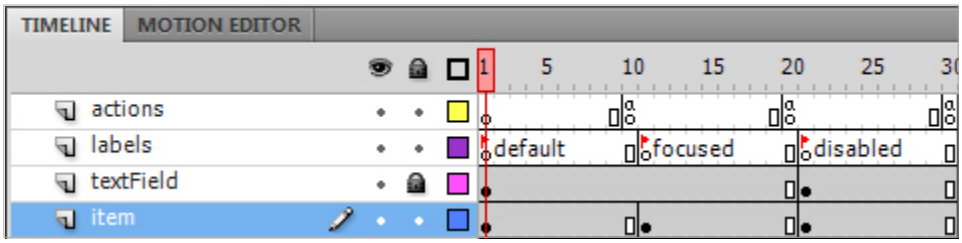


Figure 15: TextInput timeline.

2.1.4.4 Inspectable properties

TextInput .

actAsButton	true , TextInput rollover rollOut . , TextInput .
displayAsPassword	true '*' textField . textField .
Editable	false TextInput .
enabled	false .
focusable	focus / . focus false tab ,

	focus .
maxChars	textField 0 .
Text	textField .
Visible	false .

2.1.4.5

. .

- x *type*: .
- x *target*: .
- x *currentTarget*: Event .
- x *eventPhase*: . (EventPhase.CAPTURING_PHASE, EventPhase.AT_TARGET, EventPhase.BUBBLING_PHASE).
- x *bubbles*: .
- x *cancelable*: .

TextInput .

.

ComponentEvent.SHOW	true .
ComponentEvent.HIDE	false .
FocusHandlerEvent.FOCUS_IN	focus .
FocusHandlerEvent.FOCUS_OUT	focus .
Event.CHANGE	textField .
ComponentEvent.STATE_CHANGE	.
MouseEvent.ROLL_OVER	. *mouseIdx*:

(

). uint . Scaleform

MouseEventEx .

buttonIdx:

(zero base index). Scaleform

MouseEventEx .

MouseEvent.ROLL_OUT

mouseIdx:

(

). uint . Scaleform

MouseEventEx .

buttonIdx:

(zero base index). Scaleform

MouseEventEx .

textField .

```
myTextInput.addEventListener( Event.CHANGE, onTextChange);
function onTextChange(e : Event) :void {
    trace("Latest text: " + e.target.text);
}
```

2.1.5 TextArea

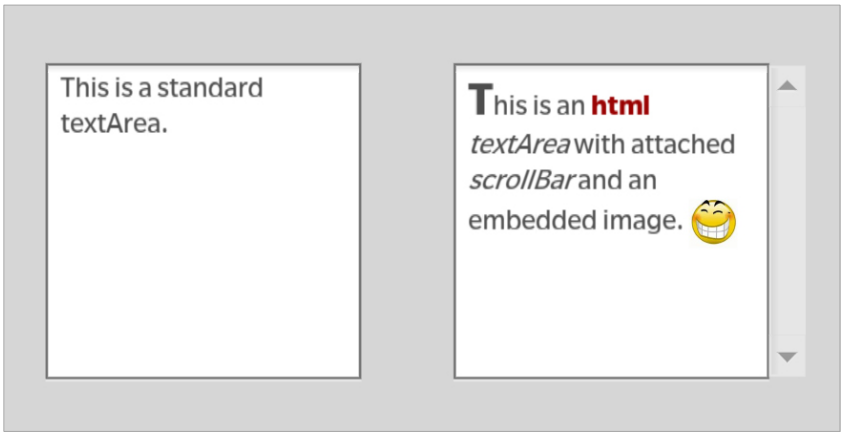


Figure 16: Unskinned TextArea.

TextArea (scaleform.clik.controls.TextArea) CLIK TextInput

(ScrollBar) . TextInput TextArea
TextInput .

Label TextInput TextArea multiline textField (wrapper) HTML ,
(wrapping), , , , textField
. TextArea textField , ,
TextArea .

2.1.5.1

TextInput textField .
.
.
.

2.1.5.2

CLIK TextArea MovieClip .

- x *textField*: TextField type.

2.1.5.3

TextInput , TextArea disabled 3
.

- x ***default*** enabled
- x ***focused*** , textField
- x ***disabled***

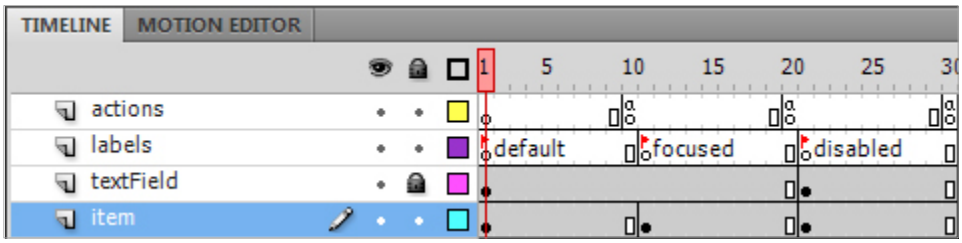


Figure 17: TextArea timeline.

2.1.5.4 Inspectable properties

TextArea TextInput password
, CLIK ScrollBar . 2.4
.

actAsButton	true focus TextInput rollOver rollOut . focus TextInput focus .
displayAsPassword	true '*' textField . textField .
editable	false TextInput .
enabled	false .
focusable	focus / . focus false tab , focus .
maxChars	textField (0)
scrollBar	CLIK ScrollBar ScrollBar ID(TextArea)
scrollPolicy	auto . on , off . ScrollBar (ScrollBar).
Text	textField .
Visible	false .

2.1.5.5

. .

- x *type*: .
- x *target*: .
- x *currentTarget*: Event .
- x *eventPhase*: . (EventPhase.CAPTURING_PHASE, EventPhase.AT_TARGET, EventPhase.BUBBLING_PHASE).
- x *bubbles*: .
- x *cancelable*:
- x .

TextArea .

.

ComponentEvent.SHOW	true .
ComponentEvent.HIDE	false .
FocusHandlerEvent.FOCUS_IN	.
FocusHandlerEvent.FOCUS_OUT	.
Event.CHANGE	textField
Event.SCROLL	.
ComponentEvent.STATE_CHANGE	.
MouseEvent.ROLL_OVER	. *mouseIdx*: ( ). uint . Scaleform MouseEventEx . *buttonIdx*: (zero base index). Scaleform MouseEventEx .
MouseEvent.ROLL_OUT	.

mouseIdx:

(

). uint . Scaleform

MouseEventEx .

buttonIdx:

(zero base index). Scaleform

MouseEventEx .

TextArea .

```
myTextArea.addEventListener( Event.SCROLL, onTextScroll);
function onTextScroll(e:Event) :void {
    // Do something
}
```

2.2 (Group Types)

RadioButton, ButtonGroup, ButtonBar . ButtonGroup Button .

. , ButtonBar Button . RadioButton ButtonGroup Button .

그림 18: *Dawn of War II*의 Radio ButtonGroup 예제

2.2.1 RadioButton

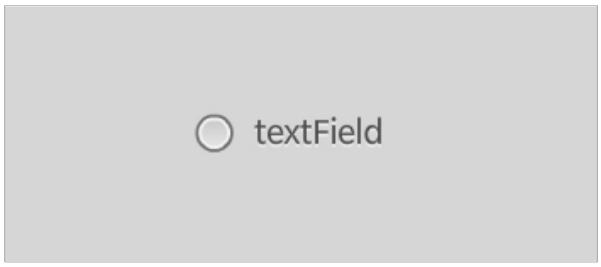


그림 19: 스킨없는 RadioButton

RadioButton (scaleform.clik.controls.RadioButton)
. RadioButton
RadioButton .

CLIK RadioButton CheckBox , , .
RadioButton , ButtonGroup(
) . RadioButton .
ButtonGroup .

2.2.1.1

RadioButton
. RadioButton , RadioButton
ButtonGroup , RadioButton .
RadioButton Button .

2.2.1.2

CLIK RadioButton MovieClip .

- x *textField*: (optional) TextField .

x *focusIndicator*: (optional) MovieClip . MovieClip .
MovieClip show hide .

2.2.1.3

CheckBox RadioButton
.

x *up*
x *over* ,
x *down*
x *disabled*
x *selected_up*
x *selected_over* , .
x *selected_ down*
x *selected_disabled*

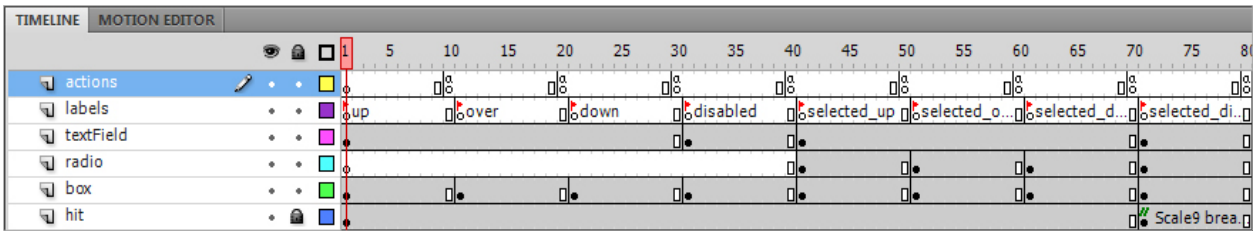


그림 20: RadioButton 타임라인.

RadioButton . Button (RadioButton
) Getting Started with CLIK
Buttons .

2.2.1.4

RadioButton Button toggle disableFocus
.

autoSize	, . autoSize autoSize=TextFieldAutoSize.NONE .
Data	. ButtonGroup .
enabled	false . .
focusable	focus / . focus false tab , focus .

groupName	RadioButton ButtonGroup . RadioButton ButtonGroup RadioButton . RadioButton _root ButtonGroup _root . RadioButton .
Label	.
selected	. . true , false ActionScript .

2.2.1.5

. .

- x *type*: .
- x *target*: .
- x *currentTarget*: Event .
- x *eventPhase*: . (EventPhase.CAPTURING_PHASE, EventPhase.AT_TARGET, EventPhase.BUBBLING_PHASE).
- x *bubbles*: .
- x *cancelable*: .

RadioButton .

.

ComponentEvent.SHOW	true .
ComponentEvent.HIDE	false .

FocusHandlerEvent.FOCUS_IN	focus .
FocusHandlerEvent.FOCUS_OUT	focus .
Event.SELECT	.
ComponentEvent.STATE_CHANGE	.
MouseEvent.ROLL_OVER	. *mouseIdx*: ( ). uint . Scaleform MouseEventEx . *buttonIdx*: (zero base index). Scaleform MouseEventEx .
MouseEvent.ROLL_OUT	. *mouseIdx*: ( ). uint . Scaleform MouseEventEx . *buttonIdx*: (zero base index). Scaleform MouseEventEx .
ButtonEvent.PRESS	. *controllerIdx*: ( ). uint . *isKeyboard:* / true false . *isRepeat*: autoRepeating

	true false .
MouseEvent.DOUBLE_CLICK	. *doubleClickEnabled* true . *mouseIdx*: ( ). uint . Scaleform MouseEventEx . *buttonIdx*: (zero base index). Scaleform MouseEventEx .
ButtonEvent.CLICK	. *controllerIdx*: ( ). uint . *isKeyboard*: / true false . *isRepeat*: autoRepeating true false .
ButtonEvent.DRAG_OVER	( ). *controllerIdx*: ( ). uint .
ButtonEvent.DRAG_OUT	( ). *controllerIdx*:

(

). uint .

ButtonEvent.RELEASE_OUTSIDE ,

.

controllerIdx.

(

). uint .0 3

RadioButton .

```
myRadio.addEventListener( Event.SELECT, onRadioToggle);
function onRadioToggle(e:Event) :void {
    // Do something
}
```

2.2.2 ButtonGroup

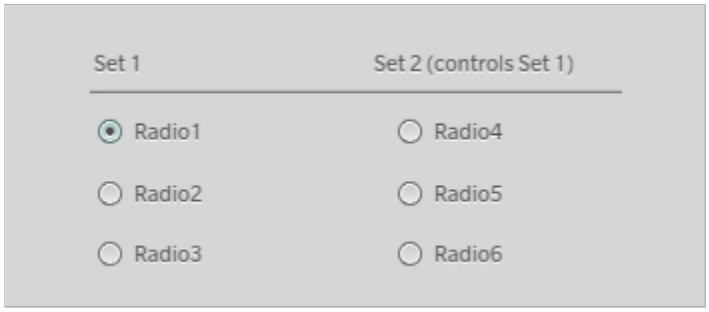


그림 21: 스킨없는 ButtonGroup.

CLIK ButtonGroup (scaleform.clik.controls.ButtonGroup)

. .

. CLIK

Button (CheckBox RadioButton) ButtonGroup

.

2.2.2.1

ButtonGroup . ButtonGroup
. RadioButtons .

2.2.2.2

CLIK ButtonGroup MovieClip .
ButtonGroup .

2.2.2.3

ButtonGroup
.

2.2.2.4

ButtonGroup
.

2.2.2.5

. .

- x *type*: .
- x *target*: .
- x *currentTarget*: Event .
- x *eventPhase*: . (EventPhase.CAPTURING_PHASE, EventPhase.AT_TARGET, EventPhase.BUBBLING_PHASE).
- x *bubbles*: .
- x *cancelable*: .

ButtonGroup .

.

Event.CHANGE	.
ButtonEvent.CLICK	. target: . CLIK Button

ButtonGroup .

```
myGroup.addEventListener( Event.CHANGE, onNewSelection);
function onNewSelection(e : Event) :void {
    if ( e. item == myRadio) {
        // Do something
    }
}
```

2.2.3 ButtonBar

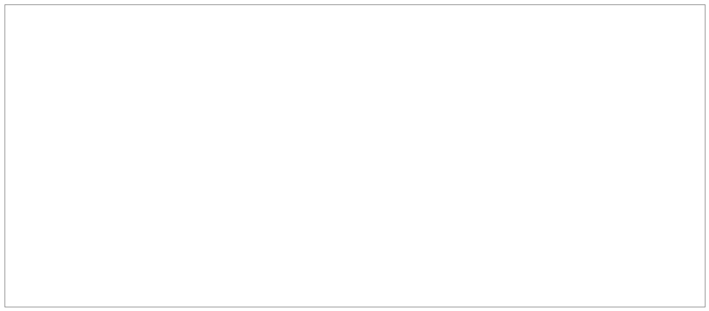


Figure 22: Unskinned ButtonBar.

ButtonBar (scaleform.clik.controls.ButtonBar) ButtonGroup ,
. , dataProvider( )
Button . ButtonBar (tab-bar) UI .
dataProvider .

```
buttonBar. dataProvider = new DataProvider([ "item1", "item2", "item3", "item4", "item5"] ) ;
```

2.2.3.1

ButtonBar ButtonGroup . ,
Button .

2.2.3.2

CLIK ButtonBar MovieClip , ButtonBar
MovieClip .

2.2.3.3

CLIK ButtonBar .
.

2.2.3.4

ButtonBar (IDE
), . ButtonBar Button
.

autoSize	true .
Direction	. , .
buttonWidth	. autoSize true

	.
enabled	false . .
focusable	focus / . focus false tab , focus .
itemRenderer	Button ID. ButtonBar .
spacing	Button . direction (direction).
Visible	false ButtonBar .

2.2.3.5

. .

- x *type*: .
- x *target*: .
- x *currentTarget*: Event .
- x *eventPhase*: . (EventPhase.CAPTURING_PHASE, EventPhase.AT_TARGET, EventPhase.BUBBLING_PHASE).
- x *bubbles*: .
- x *cancelable*: .

ButtonBar .

.

ComponentEvent.SHOW	true .
ComponentEvent.HIDE	false .
FocusHandlerEvent.FOCUS_IN	focus .

FocusHandlerEvent.FOCUS_OUT	focus .
IndexEvent.INDEX_CHANGE	. *index*: ButtonBar . int . 1 -1 ( ). *lastIndex*: ButtonBar . int . 1 -1 ( ). *data*: dataProvider . AS3 Object .
ButtonBarEvent.BUTTON_SELECT	. *index*: ButtonBar . Int . 1 -1 ( ). *renderer*: ButotnBar . .

ButtonBar Button

```
myBar.addEventListener( ButtonBarEvent.BUTTON_SELECT, onItemClick);
function onItemClick(e : ButtonBarEvent) {
    processData(e . renderer.data);
    // Do something
}
```

2.3 (Scroll Types)

ScrollIndicator, ScrollBar, Slider . ScrollIndicator
(thumb) , ScrollBar
. ScrollBar 4 ,
(thumb grip), (track), , . Slider ScrollBar
thumb track
.

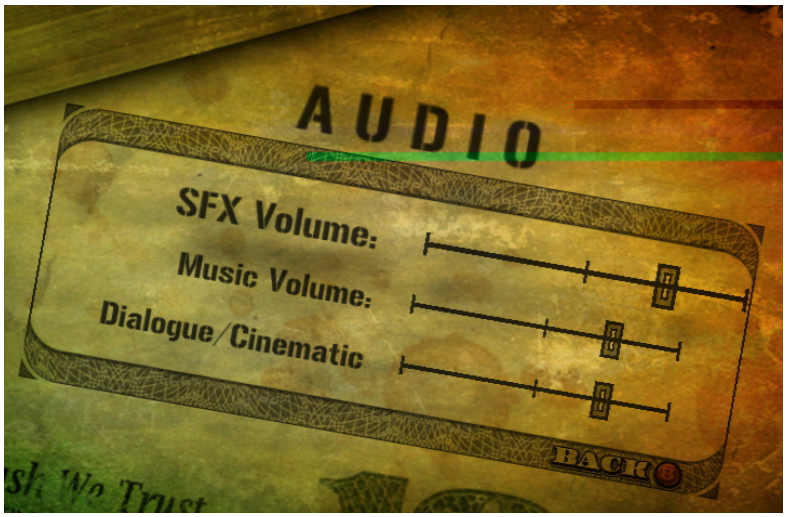


그림 23: *Mercenaries 2*의 자동 메뉴 Slider 예제

2.3.1 ScrollIndicator

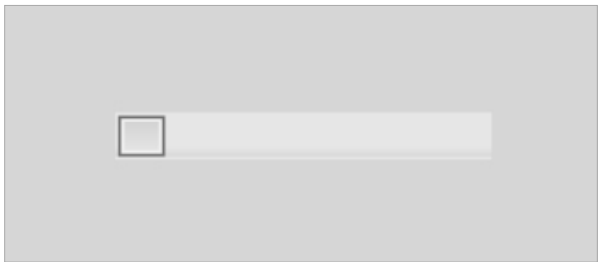


그림 24: 스킨없는 ScrollIndicator.

CLIK ScrollIndicator(scaleform.clik.controls.ScrollIndicator) multiline textField
. textField

. TextArea scrollBar ScrollIndicator ScrollBar (ID) .

2.3.1.1

ScrollInidicator . .

2.3.1.2

CLIK ScrollIndicator MovieClip .

- x ***thumb***: CLIK .
- x ***track***: MovieClip . .

.

2.3.1.3

ScrollIndicator . .

.

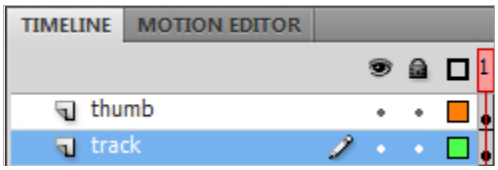


그림 25: ScrollIndicator 타임라인.

2.3.1.4

ScrollIndicator .

enabled	false . .
offsetTop	(Thumb) . .
offsetBottom	. .
scrollTarget	TextArea textField . textField ScrollIndicator .
Visible	false .

2.3.1.5

.

.

- x *type*: .
- x *target*: .
- x *currentTarget*: Event .
- x *eventPhase*: . (EventPhase.CAPTURING_PHASE, EventPhase.AT_TARGET, EventPhase.BUBBLING_PHASE).
- x *bubbles*: .
- x *cancelable*:
- x .

ScrollIndicator .

.

ComponentEvent.SHOW	true .
ComponentEvent.HIDE	false .
Event.SCROLL	ScrollIndicator .

.

```
mySI.addEventListener( Event.SCROLL, onTextScroll);
function onTextScroll(e : Event) {
    trace( " mySI.position: " + e. target.position);
    // Do something
}
```

2.3.2 ScrollBar

그림 26: 스킨없는 ScrollBar.

CLIK ScrollBar (scaleform.clik.controls.ScrollBar) .
ScrollIndicator , ,
.

2.3.2.1

ScrollBar
. ScrollBar .
2 . ,
. *trackMode* . trackMode
[SHIFT] .

2.3.2.2

CLIK ScrollBar MovieClip .

- x ***up*** *:* CLIK Button . Button; .
- x ***downArrow***: CLIK Button t . Button;
  ..
- x ***thumb***: CLIK Button .
- x ***track***: CLIK Button . (scrollbar track) (girp)
  .

2.3.2.3

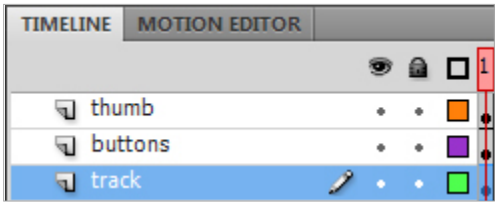


그림 27: ScrollBar timeline./ ScrollBar 타임라인

ScrollBar , . .
(thumb), up, down, .

2.3.2.4

ScrollBar ScrollIndicator , .
, trackMode:

enabled	false . .
offsetTop	(Thumb) . .

offsetBottom	. .
scrollTarget	TextArea textField . textField ScrollIndicator .
trackMode	scrollPage . scrollToCursor .
Visible	false .

2.3.2.5

. .

- x *type*: .
- x *target*: .
- x *currentTarget*: Event .
- x *eventPhase*: . (EventPhase.CAPTURING_PHASE, EventPhase.AT_TARGET, EventPhase.BUBBLING_PHASE).
- x *bubbles*: .
- x *cancelable*: .

ScrollBar .

.

ComponentEvent.SHOW	true

	.
ComponentEvent.HIDE	false .
Event.SCROLL	ScrollBar .

.

```
mySB.addEventListener( Event.SCROLL, onTextScroll);
function onTextScroll(e : Event) {
    trace( " mySB.position: " + e. target.position);
    // Do something
}
```

2.3.3 Slider

그림 28: 스킨없는 Slider.

Slider (scaleform.clik.controls.Slider) .

.

2.3.3.1

Slider Slider .

.

, [HOME], [END] .

2 . ,

. *trackMode* .

trackMode [SHIFT] .

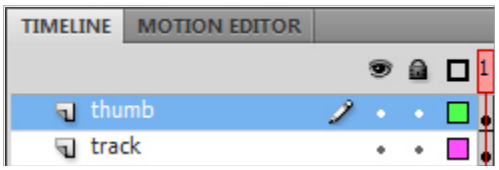
2.3.3.2

CLIK Slider MovieClip .

- x ***thumb***: CLIK Button .
- x ***track***: CLIK Button . . (girp)
  .

2.3.3.3

ScrollBar ScrollIndicator Slider (thumb, track)
.



N, 29: Slider

2.3.3.4

Slider .

Enabled	false . .
Focusable	focus / . focus false tab , focus .
Value	Slider
minimum	Slider
maximum	Slider
snapping	true snapInterval .
snapInterval	. snapping false .
liveDragging	true Slider . false .

offsetLeft	. .
offsetRight	. .
Visible	false .

2.3.3.5

. .

- x *type*: .
- x *target*: .
- x *currentTarget*: Event .
- x *eventPhase*: . (EventPhase.CAPTURING_PHASE, EventPhase.AT_TARGET, EventPhase.BUBBLING_PHASE).
- x *bubbles*: .
- x *cancelable*: .

Slider .

.

ComponentEvent.SHOW		true .
ComponentEvent.HIDE		false .
ComponentEvent.STATE_CHANGE		.
FocusHandlerEvent.FOCUS_IN	focus	.
FocusHandlerEvent.FOCUS_OUT	focus	.
SliderEvent.VALUE_CHANGE		.

Slider .

```
mySlider.addEventListener( SliderEvent.VALUE_CHANGE,
onValueChange);
function onValueChange(e : SliderEvent) {
    trace( "mySlider.value: " + e. target. value);
    // Do something
}
```

2.4 (List Types)

CLIK NumericStepper, OptionStepper, ScrollingList, TileList, DropdownMenu . NumericStepper DataProvider . ListItemRenderer , ScrollingList TileList .

NumericStepper OptionStepper . ScrollingList TileList . 2 ScrollIndicator ScrollBar . DropdownMenu (idle state) ScrollingList TileList .

그림 30: *Mercenaries 2* 의 ScrollIndicator 가 있는 ScrollingList 사용 예제

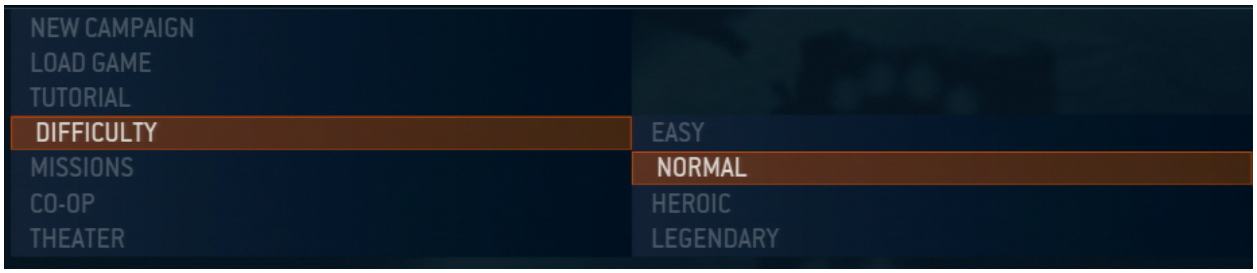


그림 31: *Halo Wars* 의 '난이도 설정' DropdownMenu 사용 예제

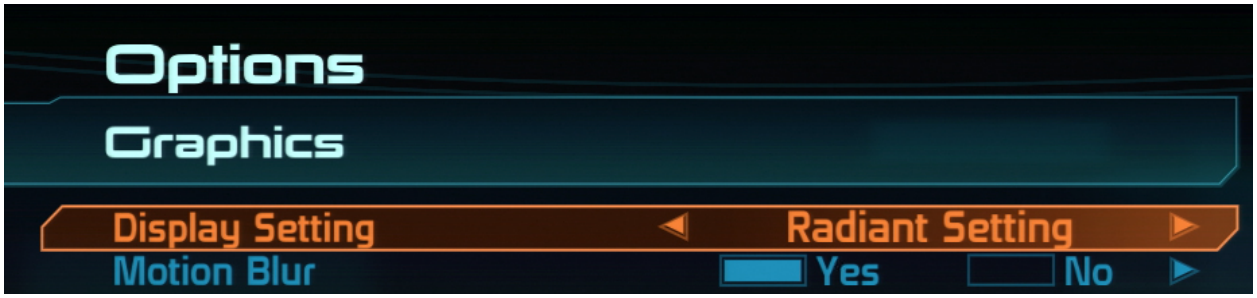


그림 32: from *Mass Effect* 의 '출력 설정' OptionStepper 사용 예제

2.4.1 NumericStepper

그림 33: 스킨없는 Numeric Stepper.

NumericStepper (scaleform.clik.controls.NumericStepper) ,

.

2.4.1.1

NumericStepper 2

.

. step . [HOME]

[END] , .

2.4.1.2

NumericStepper MovieClip

- x ***textField***: TextField . .
- x ***nextBtn***: CLIK Button . step .
- x ***prevBtn***: CLIK Button . step .

2.4.1.3

NumericStepper disabled 3 .

- x ***default*** enabled
- x ***focused*** , textField

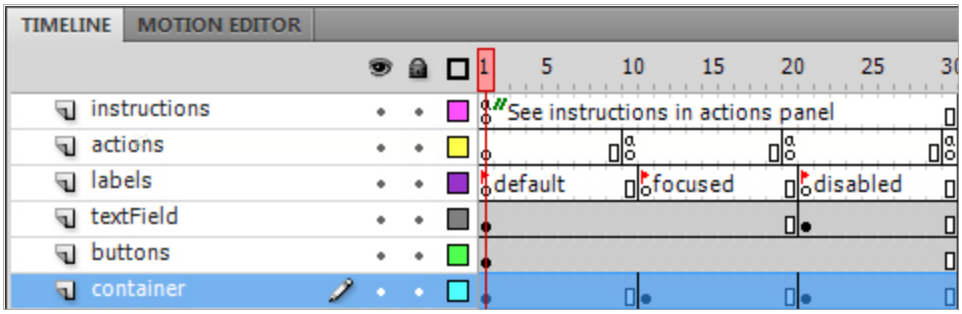


그림 34: NumericStepper 타임라인

2.4.1.4

NumericStepper MovieClip .

Enabled	false . .
Focusable	focus / . focus false tab , focus .
minimum	NumericStepper
maximum	NumericStepper
stepSize	NumericStepper NumericStepper

	/ .
Value	NumericStepper
Visible	false .

2.4.1.5

Object , .

.

- x *type*:
- x *target*:

NumericStepper .

.

ComponentEvent.SHOW	true .
ComponentEvent.HIDE	false .
ComponentEvent.STATE_CHANGE	.
FocusHandlerEvent.FOCUS_IN	focus .
FocusHandlerEvent.FOCUS_OUT	focus .
ComponentEvent.STATE_CHANGE	.
IndexEvent.INDEX_CHANGE	NumericStepper . *index*: NumericStepper . int . 1 -1 ( ). *lastIndex*: NumericStepper . int . 1 -1 ( ). *data*: dataProvider . AS3 Object .

NumericStepper

```
myNS.addEventListener( IndexEvent.INDEX_CHANGE, onValueChange);
function onValueChange(e : IndexEvent) {
    trace( " myNS.value: " + e. target. value);
    // Do something
}
```

2.4.2 OptionStepper

그림 35: 스킨없는 OptionStepper.

OptionStepper (scaleform.clik.controls.OptionStepper) NumericStepper
, . OptionStepper
dataProvider . , .
OptionStepper selectedIndex .
dataProvider hem .

```
optionStepper.dataProvider = new DataProvider([ "item1", "item2",
"item3", "item4"] ) ;
```

2.4.2.1

NumericStepper OptionStepper 2
.
.
. [HOME] [END]
dataProvider .

2.4.2.2

CLIK NumericStepper MovieClip

.

textField: TextField . .

x *nextBtn*: CLIK Button . dataProvider

.

x *prevBtn*: CLIK Button . dataProvider

.

2.4.2.3

OptionStepper disabled 3 .

x *default* enabled
x *focused* , textField
x *disabled*

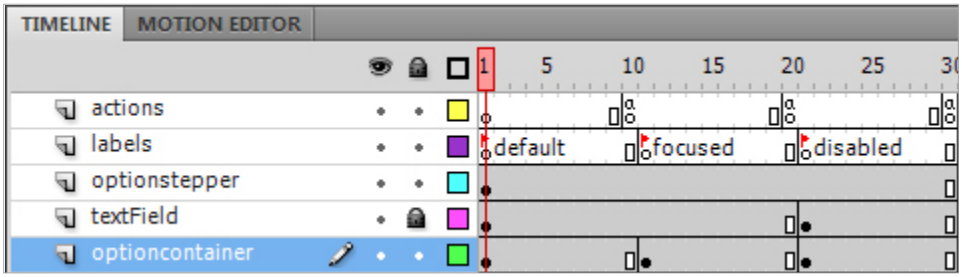


그림 36: OptionStepper 타임라인

2.4.2.4

OptionStepper MovieClip .

Enabled	false . .
Focusable	focus / . focus false tab , focus .
Visible	false .

2.4.2.5

. .

- x *type*: .
- x *target*: .
- x *currentTarget*: Event .

2.4.3 ListItemRenderer

그림 37: 스킨없는 ListItemRenderer.

ListItemRenderer (scaleform.clik.controls.ListItemRenderer) CLIK Button
Button
. , ScrollingList, TileList
DropdownMenu .

2.4.3.1

ListItemRenderer Button Button
. , ListItemRenderer
ListItemRenderer , .
ListItemRenderer .

2.4.3.2

CLIK ListItemRenderer MovieClip
.

- x ***textField***: (optional) TextField .
- x ***focusIndicator***: (optional) MovieClip . MovieClip .
  MovieClip show hide .

2.4.3.3

, ListItemRenderer
selected .

- x *up*
- x *over* ,
- x *down*
- x *disabled*
- x *selected_up*
- x *selected_over* , .
- x *selected_ down*
- x *selected_disabled*

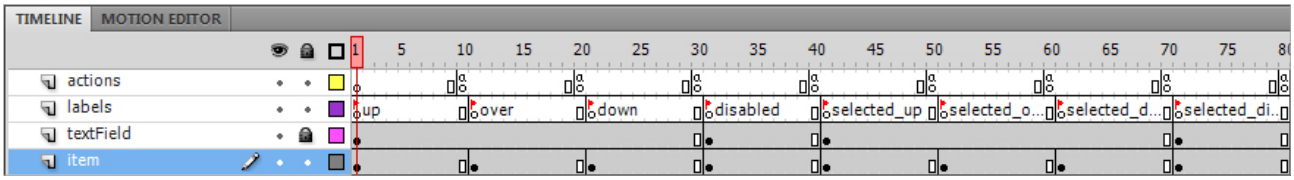


그림 38: ListItemRenderer 타임라인.

ListItemRenderer . Button (
ListItemRenderer) Getting Started with
CLIK Buttons .

2.4.3.4

ListItemRenderer ,
Button .

autoRepeat	ListItemRender "click" .
autoSize	ListItemRender , ListItemRender . autoSize autoSize=TextFieldAutoSize.NONE .
Data	ListItemRender . ListItemRender (ScrollingList/TileList) .
enabled	false . .
Label	ListItemRenderer .
Selected	ListItemRender . ListItemRender

	. ListItemRender true , false ActionScript .
Toggle	ListItemRender . true ListItemRender .
Visible	false .

2.4.3.5

. .

- x *type*: .
- x *target*: .
- x *currentTarget*: Event .
- x *eventPhase*: . (EventPhase.CAPTURING_PHASE, EventPhase.AT_TARGET, EventPhase.BUBBLING_PHASE).
- x *bubbles*: .
- x *cancelable*: .

ListItemRenderer .

.

ListItemRenderer

. ListItemRenderer

ITEM_PRESS ITEM_CLICK ListEvent . EventListener

ListItemRenderer EventListener

(ListEvent.ITEM_CLICK) .

ComponentEvent.SHOW	true .
ComponentEvent.HIDE	false .
ComponentEvent.STATE_CHANGE	.
FocusHandlerEvent.FOCUS_IN	focus .
FocusHandlerEvent.FOCUS_OUT	focus .
Event.SELECT	.
MouseEvent.ROLL_OVER	. *mouseIdx*: ( ). uint . Scaleform MouseEventEx . *buttonIdx*: (zero base index). Scaleform MouseEventEx .
MouseEvent.ROLL_OUT	. *mouseIdx*: ( ). uint . Scaleform MouseEventEx . *buttonIdx*: (zero base index). Scaleform MouseEventEx .
ButtonEvent.PRESS	. *controllerIdx*: ( ). uint . *isKeyboard:* / true

	false .
	isRepeat: autoRepeating
	true
	false .
MouseEvent.DOUBLE_CLICK	. *doubleClickEnabled*
	true .
	mouseIdx:
	(
	). uint . Scaleform
	MouseEventEx .
	buttonIdx:
	(zero base index). Scaleform
	MouseEventEx .
ButtonEvent.CLICK	.
	controllerIdx:
	(
	). uint .
	isKeyboard: /
	true
	false .
	isRepeat: autoRepeating
	true
	false .
ButtonEvent.DRAG_OVER	(
	).
	controllerIdx:
	(
	). uint .
ButtonEvent.DRAG_OUT	(

	). *controllerIdx*. ( ). uint .
ButtonEvent.RELEASE_OUTSIDE	, . *controllerIdx*. ( ). uint .

2.4.4 ScrollingList

그림 39: 스킨없는 ScrollingList.

ScrollingList (scaleform.clik.controls.ScrollingList) .

, . ScrollIndicator ScrollBar . dataProvider , dataProvider .

```
scrollingList.dataProvider = new DataProvider([ "item1", "item2", "item3", "item4"]);
```

ScrollingList ListItemRenderer . ListItemRenderer FLA . itemRenderer . .

2.4.4.1

ScrollBar ScrollingList .

. .

ScrollingList .

ScrollingList *wrapping* (

). *wrapping* normal

. , *wrapping* wrap

wrap . *wrapping* stick

.

[Page Up] [Page Down] .

[Home] [End] ,

. ScrollBar ScrollingList .

ScrollBar .

. ,

D- . CLIK UI

.

2.4.4.2

ScrollingList . ScrollingList

.

2.4.4.3

ScrollingList disabled 3 .

- x ***default*** enabled
- x ***focused*** ,

x *disabled*

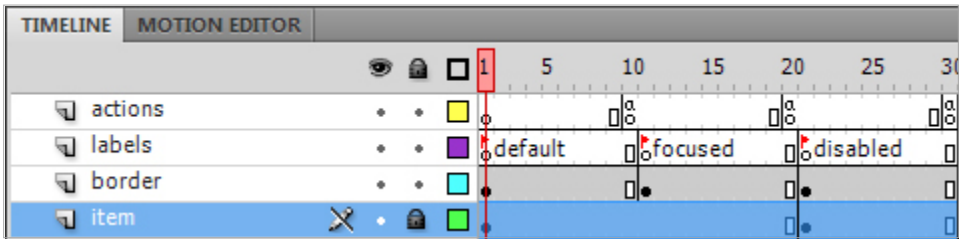


그림 40: ScrollingList 타임라인

2.4.4.4

ScrollingList MovieClip .

Enabled	false . .
Focusable	focus / . focus false tab , focus .
itemRenderer	ListItemRenderer .

	. *itemRendererInstanceName* (, ListItemRenderer).
itemRendererInstanceName	ScrollingList (prefix). . 'r' 'r1', 'r2', 'r3',... . 1 .
Margin	. item*rendererInstanceName* .
padding	. itemRendererInstanceName (, ListItemRenderer). ScrollBar .
rowHeight	. itemRendererInstanceName (, ListItemRenderer).
Visible	false . ScrollBar ListItemRenderer .

2.4.4.5

. .

x *type*: .

- x *target*: .
- x *currentTarget*: Event .
- x *eventPhase*: . (EventPhase.CAPTURING_PHASE, EventPhase.AT_TARGET, EventPhase.BUBBLING_PHASE).
- x *bubbles*: .
- x *cancelable*:
- x .

ScrollingList .

.

ComponentEvent.SHOW	true .
ComponentEvent.HIDE	false .
ComponentEvent.STATE_CHANGE	.
FocusHandlerEvent.FOCUS_IN	focus .
FocusHandlerEvent.FOCUS_OUT	focus .
ListEvent.INDEX_CHANGE	. x *itemRenderer*: . IListItemRenderer . x *itemData*: . dataProvider . ActionScript . x *index*: . int . 1 -1 (selectedIndex). x *controllerIdx*: / ( ).
ListEvent.ITEM_PRESS	

	x *itemRenderer*: . IListItemRenderer . x *itemData*: . dataProvider . ActionScript . x *index*: . int . 1 -1 (selectedIndex). x *controllerIdx*: / ( ).
ListEvent.ITEM_CLICK	. x *itemRenderer*: . IListItemRenderer . x *itemData*: . dataProvider . ActionScript . x *index*: . int . 1 -1 (selectedIndex). x *controllerIdx*: / ( ).
ListEvent.ITEM_DOUBLE_CLICK	. x *itemRenderer*: . IListItemRenderer . x *itemData*: .

	dataProvider . ActionScript . x *index*: . int . 1 -1 (selectedIndex). x *controllerIdx*: / ( ).
ListEvent.ITEM_ROLL_OVER	. *itemRenderer*: . IListItemRenderer . x *itemData*: . dataProvider . ActionScript . x *index*: . int . 1 -1 (selectedIndex). x *controllerIdx*: / ( ).
ListEvent.ITEM_ROLL_OUT	The mouse cursor has rolled out of a list item. x *itemRenderer*: . IListItemRenderer . x *itemData*: . dataProvider . ActionScript . x *index*: . int

. 1 -1

(selectedIndex).

x *controllerIdx*.

/ (

).

.

```
myList.addEventListener(ListEvent.ITEM_CLICK, onItemClicked);
function onItemClicked(e : ListEvent) {
    trace( " ListItemRenderer Clicked: " + e. itemRenderer);
    // Do something
}
```

2.4.5 TileList

TileList (scaleform.clik.controls.TileList) ScrollingList

. ,

. ScrollIndicator ScrollBar

. TileList ScrollingList

TileList . 4

. dataProvider , dataProvider

.

```
tileList.dataProvider = new DataProvider([ "item1", "item2",
"item3", "item4", "item5"] ) ;
```

TileList ListItemRenderer . ListItemRenderer FLA

. itemRenderer

. .

2.4.5.1

ScrollBar TileList .

. ,

. TileList

4 .

. TileList

.

[Page Up] [Page Down] .

[Home] [End] ,

. ScrollBar ScrollingList .

ScrollBar .

2.4.5.2

TileList . TileList

.

2.4.5.3

TileList disabled 3 .

- x ***default*** enabled
- x ***focused*** ,
- x ***disabled***

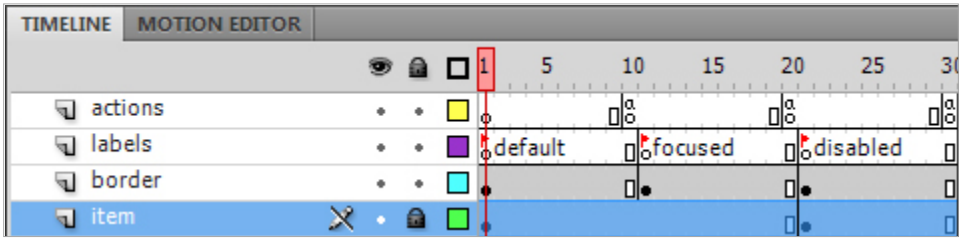


그림 41: TileList 타임라인

2.4.5.4

TileList MovieClip .

columnWidth	. *rendererInstanceName* .
Direction	. .
enabled	Disables the component if set to false. Disabled components will no longer receive user input.
externalColumnCount	*rendererInstanceName* TileList .
focusable	Enable/disable focus management for the component. Setting the focusable property to false will remove support for tab key, direction key and mouse based focus changes.
itemRenderer	ListItemRenderer . .

	itemR*endererInstanceName* (, ListItemRenderer).
itemRendererInstanceName	ScrollingList . . 'r' . 'r1', 'r2', 'r3',... 1 .
Margin	. *itemRendererInstanceName* (, ListItemRenderer).
Padding	. itemRendererInstanceName (, ListItemRenderer). ScrollBar .
rowHeight	. itemR*endererInstanceName* (, ListItemRenderer).
Visible	false . ScrollBar ListItemRenderer .

2.4.5.5

. .

x *type*: .

- x *target*: .
- x *currentTarget*: Event .
- x *eventPhase*: . (EventPhase.CAPTURING_PHASE, EventPhase.AT_TARGET, EventPhase.BUBBLING_PHASE).
- x *bubbles*: .
- x *cancelable*: .

TileList .

.

ComponentEvent.SHOW	true .
ComponentEvent.HIDE	false .
ComponentEvent.STATE_CHANGE	.
FocusHandlerEvent.FOCUS_IN	focus .
FocusHandlerEvent.FOCUS_OUT	The component has lost focus.
ListEvent.INDEX_CHANGE	. x *itemRenderer*: . IListItemRenderer . x *itemData*: . dataProvider . ActionScript . x *index*: . int . 1 -1 (selectedIndex). x *controllerIdx*: / ( ).

ListEvent.ITEM_PRESS	. x *itemRenderer*: . IListItemRenderer . x *itemData*: . dataProvider . ActionScript . x *index*: . int . 1 -1 (selectedIndex). x *controllerIdx*: / ( ).
ListEvent.ITEM_CLICK	. x *itemRenderer*: . IListItemRenderer . x *itemData*: . dataProvider . ActionScript . x *index*: . int . 1 -1 (selectedIndex). x *controllerIdx*: / ( ).
ListEvent.ITEM_DOUBLE_CLICK	. x *itemRenderer*: . IListItemRenderer .

	x *itemData*: . dataProvider . ActionScript .
	x *index*: . int . 1 -1 (selectedIndex).
	x *controllerIdx*: / ( ).
ListEvent.ITEM_ROLL_OVER	.
	x *itemRenderer*: . IListItemRenderer .
	x *itemData*: . dataProvider . ActionScript .
	x *index*: . int . 1 -1 (selectedIndex).
	x *controllerIdx*: / ( ).
ListEvent.ITEM_ROLL_OUT	.
	x *itemRenderer*: . IListItemRenderer .
	x *itemData*: . dataProvider .

ActionScript .

x *index*. . int

. 1 -1

(selectedIndex).

x *controllerIdx*.

/ (

).

TileList .

```
myList.addEventListener( FocusHandlerEvent.FOCUS_IN,
onListFocused);
function onListFocused(e : FocusHandlerEvent) {
trace( " myList is now focused!");
// Do something
}
```

2.4.6 DropdownMenu

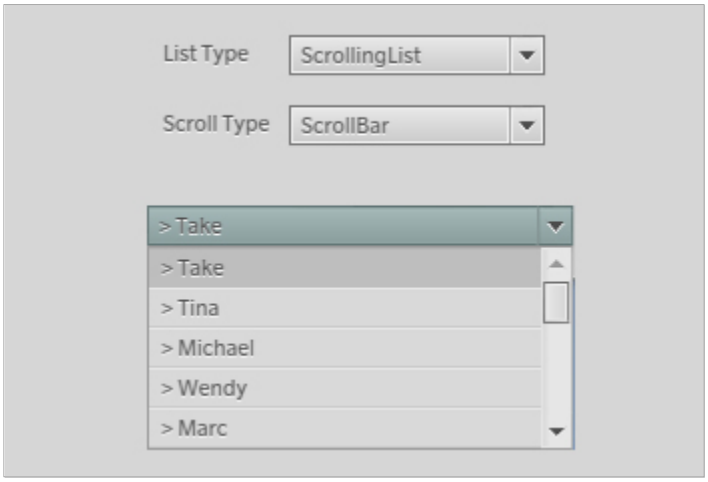


그림 42: 스킨없는 DropdownMenu.

DropdownMenu (scaleform.clik.controls.DropdownMenu) .

. DropdownMenu (idle state)

. ScrollingList TileList , ScrollBar

ScrollIndicator . DropdownMenu dataProvider
, dataProvider .

```
dropdownMenu. dataProvider = new DataProvider([ "item1", "item2",
"item3", "item4"]);
```

DropdownMenu ScrollingList .
ListItemRenderer FLA . dropdown
. .

, DropdownMenu scrollbar . Scrollbar
ScrollIndicator . .

2.4.6.1

DropdownMenu [Enter]
. . ScrollingList,
TileList, ScrollBar .
, DropdownMenu .
DropdownMenu .

2.4.6.2

DropdownMenu Button .
DropdownMenu MovieClip .
.

- x ***textField***: (optional) TextField .
- x ***focusIndicator***: (optional) MovieClip . MovieClip .
  MovieClip show hide .

2.4.6.3

DropdownMenu . ToggleButton CheckBox
.

- x *up*
- x *over* ,
- x *down*
- x *disabled*
- x *selected_up*
- x *selected_over* , .
- x *selected_ down*
- x *selected_disabled*

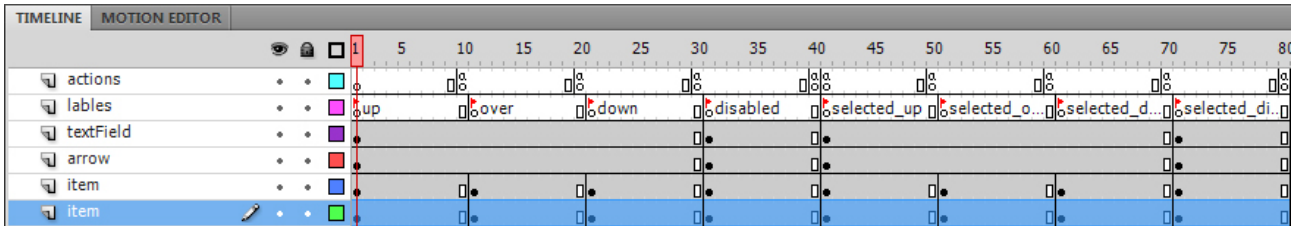


그림 43: DropdownMenu 타임라인

DropdownMenu . Button (DropdownMenu) Getting Started with CLIK Buttons .

2.4.6.4

DropdownMenu .

autoSize	DropdownMenu , . autoSize autoSize="none" .
Dropdown	DropdownMenu ((ScrollingList TileList)
Enabled	false .
Focusable	focus . false focus .
menuDirection	. "up" "down" .

menuMargin	. .
menuOffset	. (+) . (+) .
menuPadding	, , . scrollbar .
menuRowCount	.
menuWidth	If set, this number will be enforced as the width of the drop down's list.
thumbOffset	Scrollbar . scrollbar .
Scrollbar	. . .
Visible	false

2.4.6.5

. .

- x *type*: .
- x *target*: .
- x *currentTarget*: Event .
- x *eventPhase*: . (EventPhase.CAPTURING_PHASE, EventPhase.AT_TARGET, EventPhase.BUBBLING_PHASE).

x *bubbles*: .

x *cancelable*: .

DropdownMenu change Button

. .

ComponentEvent.SHOW	true .
ComponentEvent.HIDE	false .
ComponentEvent.STATE_CHANGE	.
Event.SELECT	.
FocusHandlerEvent.FOCUS_IN	focus .
FocusHandlerEvent.FOCUS_OUT	focus .
ListEvent.INDEX_CHANGE	. x *itemRenderer*: . IListItemRenderer . x *itemData*: . dataProvider . ActionScript . x *index*: . int . 1 -1 (selectedIndex). x *controllerIdx*: / ( ).
MouseEvent.ROLL_OVER	. *mouseIdx*: ( ). uint . Scaleform , MouseEventEx .

	buttonIdx: (zero base index). Scaleform , MouseEventEx .
MouseEvent.ROLL_OUT	. *mouseIdx*: uint ( ). . Scaleform , MouseEventEx . *buttonIdx*: (zero base index). Scaleform , MouseEventEx .
ButtonEvent.PRESS	DropdownMenu . *controllerIdx*: ( ). uint . *isKeyboard:* / true false . *isRepeat*: autoRepeating true false .
MouseEvent.DOUBLE_CLICK	. *doubleClickEnabled* **프로퍼티가 true** . *mouseIdx*: ( ). uint . Scaleform , MouseEventEx . *buttonIdx*:

	(zero base index). Scaleform , MouseEventEx .
ButtonEvent.CLICK	DropdownMenu . *controllerIdx*: ( ). uint . *isKeyboard:* / true false . *isRepeat*: autoRepeating true false .
ButtonEvent.DRAG_OVER	DropdownMenu ( ). *controllerIdx*: ( ). uint .
ButtonEvent.DRAG_OUT	DropdownMenu ( ). *controllerIdx*: ( ). uint .
ButtonEvent.RELEASE_OUTSIDE	. *controllerIdx*: ( ). uint .

2.5 (Progress Types)

.

CLIK 2 : StatusIndicator Progress Bar.
StatusIndicator , ProgressBar StatusIndicator
, .

그림 44: *Mercenaries 2*의 팩션 StatusIndicator 사용예

2.5.1 StatusIndicator

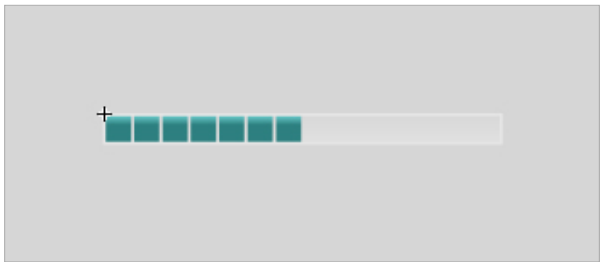


그림 45: 스킨없는 StatusIndicator.

StatusIndicator (scaleform.clik.controls.StatusIndicator)
(indicator) . StatusIndicator
.

.

2.5.1.1

StatusIndicator .

2.5.1.2

StatusIndicator MovieClip . , StatusIndicator
2 . stop()
. StatusIndicator value gotoAndStop()
.

2.5.1.3

StatusIndicator .
.

2.5.1.4

StatusIndicator MovieClip .

Visible	false .
Enabled	true disable .
Value	. .

Minimum
Maximum

2.5.1.5

. .

- x *type*: .
- x *target*: .
- x *currentTarget*: Event .
- x *eventPhase*: . (EventPhase.CAPTURING_PHASE, EventPhase.AT_TARGET, EventPhase.BUBBLING_PHASE).
- x *bubbles*: .
- x *cancelable*: .

StatusIndicator .

.

ComponentEvent.SHOW	true .
ComponentEvent.HIDE	false .

2.6 (Other Types)

CLIK .

modal modal-free

.

DragSlot UI CLIK

. CLIK CLIK DragManager

(scaleform.clik.managers.DragManager.as).

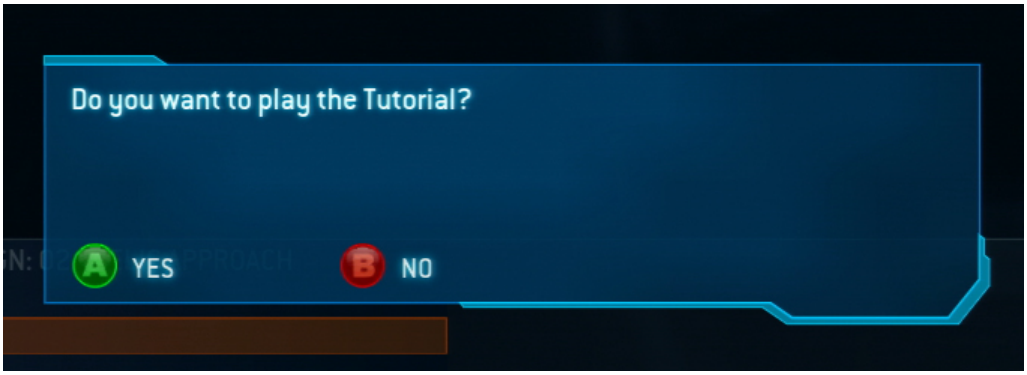


Figure 46: Window example from Halo Wars **.**

2.6.1 Window

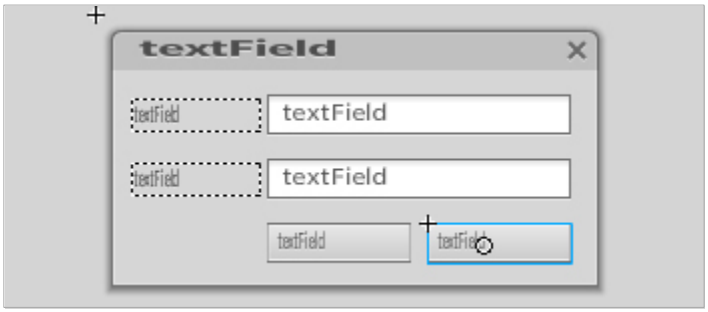


Figure 47: Sample unskinned Window.

CLIK (scaleform.clik.controls.Window)

. , .

PopUpManager (PopUpManager.showModal())

.

2.6.1.1

Window .

2.6.1.2

Window MovieClip .

- x *closeBtn*: ( ) CLIK . .
- x *titleBtn*: ( ) CLIK . .
- x *okBtn*: ( ) CLIK . "Accept" "OK" .
- x *resizeBtn*: ( ) CLIK .

.

- x *background* ( ) MovieClip . .

.

- x *hit*: ( ) MovieClip . hitArea .

2.6.1.3

Dialog . MovieClip ,

.

2.6.1.4

Window MovieClip .

contentPadding	, , .

Enabled	false .
maxHeight	.
maxWidth	.
minHeight	.
minWidth	.
Source	.
Title	titleBtn , .
Visible	false .

2.6.1.5

. .

- x *type*: .
- x *target*: .
- x *currentTarget*: Event .
- x *eventPhase*: . (EventPhase.CAPTURING_PHASE, EventPhase.AT_TARGET, EventPhase.BUBBLING_PHASE).
- x *bubbles*: .
- x *cancelable*: .

Window .

.

ComponentEvent.SHOW	true .
ComponentEvent.HIDE	false .

(submit)

```
myWindow. addEventListener(  ComponentEvent.HIDE,  onWindowClosed);
function  onWindowClosed(  e: ComponentEvent) {
        // Do something.
}
```

3 아트 세부설명

CLIK .

, .

3.1

CLIK .

3.1.1 픽셀퍼펙트 이미지(Pixel-perfect Images)

CLIK .

.

1. View
2. Grid → Show Grid
3. View
4. Grid → Edit Grid
5. , 1px
6. OK

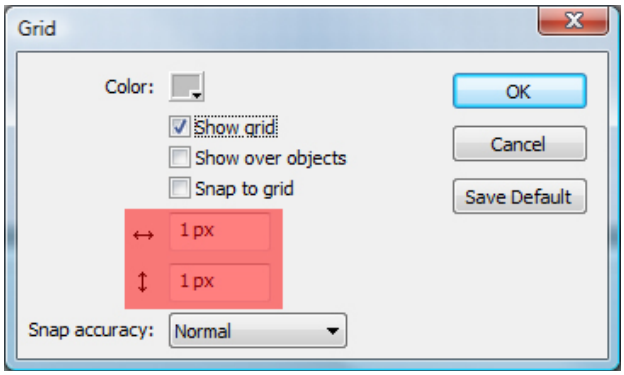


그림 48: Edit Grid 창.

. 1 .

. SWF (blur) . :

2 . .

"2 "

.

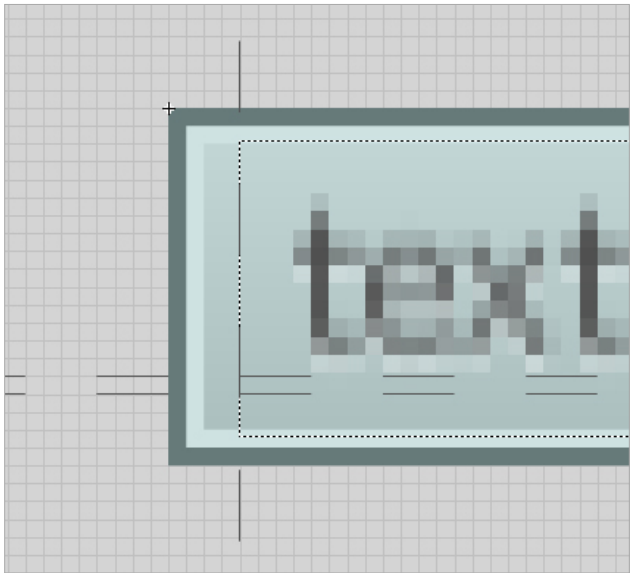


그림 49: 픽셀퍼펙트 벡터 그래픽 스킨

3.1.2 마스크(Masks)

.

. , . ,

.

(single dsigit) .

PNG

. , .

3.1.3 애니메이션(Animations)

.

. ,

.

tessellation( )

. tessellation .

tween tween

tween . tween

.

3.1.4 레이어와 그리기 기본요소(Layers and Draw Primitives)

.

.

.

3.1.5 복잡한 스킨

.

(blurring) .

3.1.6 2 의 제곱

2 . 2 .

- X 16x16
- X 32x32
- X 64x64
- X 128x128

- X 128x256
- X 256x256
- X 512x256
- X 512x512

3.2

CLIK

.

3.2.1 컴포넌트 복제

. 2

.

3.2.1.1 (1)

( ) CLIK (Button) FLA

FLA .

1. *CLIK_Components.fla* .
2. (Button) FLA .
   [→ *Copy*], FLA [→
   Paste] .
3. FLA *Rename* 'Button'
   .
4. FLA *Properties* .
5. *Identifier* 3 .
6. *Paste* .
   .

3.2.1.2 (2)

( )

. .

1. *Properties* .
2. *Properties* (: scaleform.clik.controlsButton) *Class*
   . [CTRL+C] .
3. *Cancel* .
4. *Duplicate* .
5. *Duplicate Symbol* *Export for ActionScript* .
6. *Class* [CTRL+V] .
7. *OK*
8. *Component Definition* .
9. *Class* [CTRL+V] .
10. *OK*

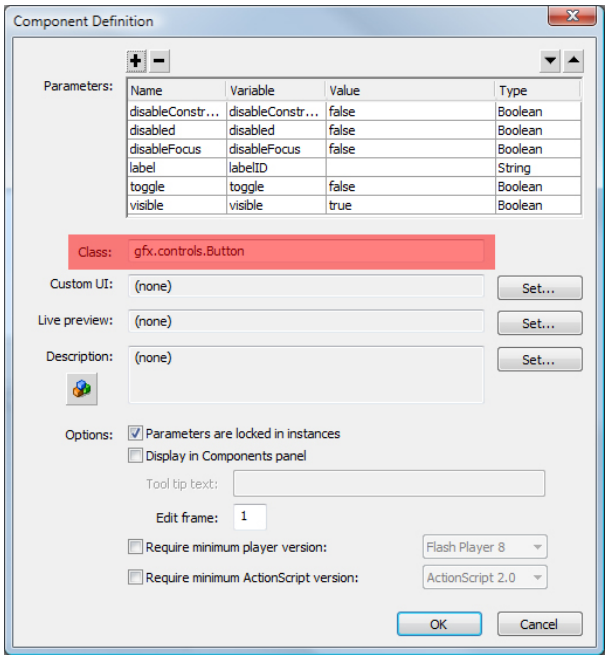


그림 51: 컴포넌트 정의 (클래스 부분을 채워 넣어야 한다).

3.3

CLIK .

. (look and feel)

.

CLIK ,

.

.

CLIK . FLA

,

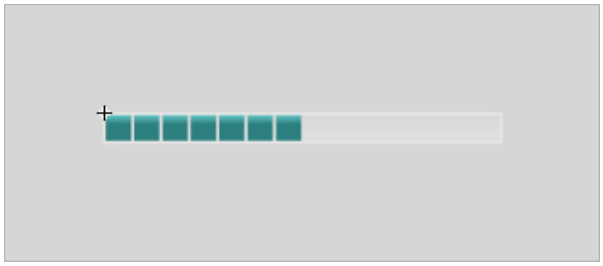
x

x .

x

Getting Started with CLIK .

3.3.1 StatusIndicator 스키닝



52: StatusIndicator

StatusIndicator . StatusIndicator
. *indicator* *track* 2 . *track*
. *indicator* ,
.

,
. StatusIndicator
. ,
StatusIndicator .

그림 53: StatusIndicator PSD 파일

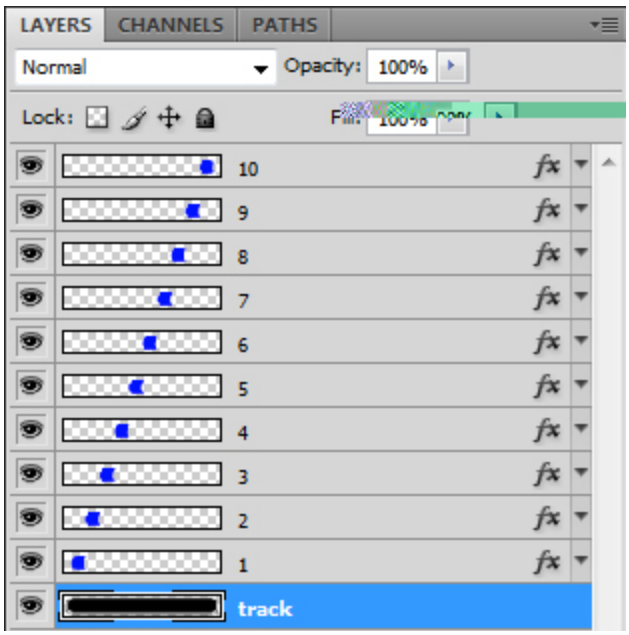


그림 54: StatusIndicator PSD 파일의 포토샵 레이어 설정

1. StatusIndicator .

   . .

2. PSD .
3. *File → Import → Import to Stage*
4. StatusIndicator PSD
5. *Import* *Convert layers to* Layers
6. *OK*
7. PSD .

   10 , PSD 10 (

). 1 10 , 1

   , 10 . 1 .
8. .
9. . .
10. 1 6 .
11. 1 11 → *Insert Keyframe*
12. 2 [CTRL+X] .
13. 1 11 [CTRL+SHIFT+V] 1

    .
14. 10 (1)

    . 5 .

2 11, 3 16, 4
21 . 10 PSD , 51
.

15. 2~10 .
16. , 5
, → *Remove Frames*
. 55 .
17. *indicator* .
18. *track* . *track* .
19. 1 indicator .
20. indicator track actions . , track
indicator actions

3.4.1 개요

Scaleform ( ) SWF
ScaleformLocalization System . UI
.

3.4.2 폰트 내장

Scaleform .
Scaleform
. Scaleform
. Slate Mobile textField .
Slate Mobile , , (CJK)
ASCII . Slate Mobile CJK
, .

textField Scaleform Localization System (3.4.4)
. Scaleform Localization System
. Localization
. UI
localization .

3.4.3 textField 에 폰트 내장

, textField .
( ) . textField textField
(Window → Property Inspector) *Embed* .
OK . FLA
. FLA textField
. ,
.
.

3.4.4 Scaleform Localization System

Scaleform Localization System hot-swapping
/ . SWF
SWF . Scaleform
.

Scaleform Localization System textField
textField import .
gfxfontlib.fla SWF import . import
textField . Scaleform
.

export . (
) . Scaleform Localization System fontconfig.txt
textField
.

, fontconfig.txt . Scaleform
textField . textField
'$TITLE' '$TITLE=Scaleform'
textField 'Scaleform' .

Scaleform Localization System .
Scaleform Localization System Font Overview
.

4 프로그래밍 세부사항

, CLIK

.

그림 57: 스케일폼 CLIK 클래스 계층구조

4.1 UIComponent

CLIK

. UIComponent(scaleform.clik.core.UIComponent)

. CLIK ,
UIComponent .

UIComponent 8 MovieClip .
MovieClip . UIComponent /
.

- x enabled;
- x visible;
- x focused;
- x focusable;
- x width;
- x height;
- x scaleX;
- x scaleY;
- x displayFocus treu .
  " "

UIComponent .

- x *configUI*: .
- x *draw*: . " "
- x *changeFocus*:
- x *scrollWheel*: .

UIComponent EventDispatcher . ,
. .

4.1.1 초기화

onLoad() .

- x MovieClip
- x . configUI()
- x . validateNow() . ,
  draw()

4.2

CLIK .

.

.

4.2.1 Button 컴포넌트

( ) .

CLIK . ListItemRenderer, RadioButton, CheckBox . ScrollBar Button . CLIK Button (scaleform.clik.controls.Button) Button .

.

- x *up*
- x *over* ,
- x *down*
- x *disabled* disable

Button (prefixed state) . . CLIK Button "selected_" ,

.

(selected) Button .

- x *up*
- x *over* ,
- x *down*
- x *disabled* disable
- x *selected_up*
- x *selected_over* ,
- x *selected_ down*
- x *selected_disabled*

주의: CLIK Button .

Getting Started with CLIK Buttons .

CLIK getStatePrefixes() ,

. .

```
protected function getStatePrefixes():Vector.<String> {
    return _selected ? statesSelected : statesDefault;
}
```

CLIK Button "selected_" .

getStatePrefixes() .

.

, ( )

. Button stateMap . CLIK

.

```
protected var _stateMap:Object = {
        up:["up"],
        over:["over"],
        down:["down"],
        release: ["release", "over"],
        out:["out","up"],
        disabled:["disabled"],
        selecting: ["selecting", "over"],
        toggle: ["toggle", "up"],
        kb_selecting: ["kb_selecting", "up"],
        kb_release: ["kb_release", "out", "up"],
        kb_down: ["kb_down", "down"]
    }
```

.

getStatePrefixes() .

.

Figure 58: Process used to determine the correct keyframe to play.

. ,

.

.

4.2.2 버튼이 아닌 상호작용 컴포넌트

, .

CLIK ScrollingList, OptionStepper, Slider, TextArea .

.

.

- x *default*
- x *focused*
- x *disabled*

4.2.3 비 상호작용 컴포넌트

, disable .

.

- x *default*
- x *disabled*

4.2.4 특수한 경우

. StatusIndicator

ProgressBar .

(%) . 0, 10 50

StatusIndicator , 5(50%) gotoAndStop()

25(50 50%) .
. updateValue() ,
.

,
. TextInput TextArea asAsButton
roll over roll out over out
.

4.3

CLIK .
" (dispatch)" . ,
. .
ActionScript 3 CLIK .

ActionScript 3 .
http://help.adobe.com/en_US/FlashPlatform/reference/actionscript/3/flash/events/Event.html

4.3.1 사용예와 가장 좋은 연습(Usage and Best Practices)

4.3.1.1 이벤트에 신청하기(Subscribing to an Event)

addEventListener() .
. removeEventListener() .
.
.

CLIK
. CLIK
scaleform.clik.events .

addEventListener() 3 . **useCapture, priority**

useWeakReference . ,

.

addEventListener() .

```
        buttonInstance.addEventListener(ButtonEvent.CLICK, onButtonClick, false,
0, true);
```

addEventListener() .

http://help.adobe.com/en_US/FlashPlatform/reference/actionscript/3/flash/events/EventDispatcher.html#addEventListener()

```
        buttonInstance.addEventListener(ButtonEvent.CLICK, onButtonClick, false,
0, true);
        function      onButtonClick (e:Event):void              {
                buttonInstance.removeEventListener(             ButtonEvent.CLICK,
onButtonClick, false);
        }
```

buttonInstance ButtonEvent.CLICK

. onButtonClick .

.

.

CLIK ( ) [

] .

4.3.1.2 이벤트 발송(Dispatching an Event)

CLIK

dispatchEvent() . .

, .

,

. .

```
        dispatchEvent(new Event(Event.CHANGE));
```

4.4

ActionScript 3 .

```
import flash.utils.getDefinitionByName;
import scaleform.clik.controls.Button;

public var myButton:Button;

// Retrieve a reference to the Class definition.
var classRef:Class = getDefinitionByName("ButtonLinkageID") as Class;
// Instantiate a new instance of the class and store it.
if (classRef != null) { myButton = new classRef () as Button; }
myButton.addEventListener(ButtonEvent.CLICK, onButtonClick, false, 0, true);
```

"ButtonLinkageID" scaleform.clik.controls.Button

.FLA .

BaseClass .FLA ,

.

4.5

CLIK (

), .

FocusHandler (scaleform.clik.managers.FocusHandler)

. FocusHandler .

focus Scaleform ( ) focus

ActionScript . focus ActionScript .

- myUIComponent.focused = 1; // 1 0 .
- stage.focus = myMovieClip; // MovieClip .
- FocusHandler.getInstance().setFocus(myMovieClip); // FocusHandler .

4.5.1 사용예와 가장 좋은 연습(Usage and Best Practices)

focus InteractiveDisplayObject , focus

.

(,). true .

.

false ***tabEnabled*** ***mouseEnabled*** MovieClip Scaleform Flash

focus focus . CLIK

tabEnabled mouseEnabled .

tabEnabled mouseEnabled

focus . AS3 MovieClip.focusEnabled

UIComponent.***focusable*** . false

UIComponent CLIK focus focus . ***focusable***

tabEnabled mouseEnabled

tabEnabled mouseEnabled .

(ScrollBar) focusTarget

. FocusHandler

focusTarget focusTarget .

. displayFocus true .

Slider Slider .

UIComponent.changeFocus() .

```
function changeFocus():void {
    track.displayFocus = _focused;
}
```

,

focus .

tabEnabled false .

```
background. tabEnabled = false;
```

4.5.2 복합 컴포넌트에서 포커스 캡처

, ScrollingList,

OptionStepper, ButtonBar .

, . , Scaleform

, ,

.

.

1. tabEnabled = mouseEnabled = focusable = false
   (, OptionStepper)
2. focusTarget .
   focusable = true .
3. changeFocus displayFocus true .

, , .

4.6

CLIK 8 MovieClip

MovieClip .

. , CLIK

.

4.6.1 사용예와 가장 좋은 연습(Usage and Best Practices)

InputDelegate(scaleform.clik.managers.InputDelegate)

. InputDelegate InputDetails(scaleform.clik.ui.InputDetails) ,

. InputDelegate.readInput()
. InputDetails InputDelegate .

InputDetails .

- x "key"
- x
- x . Key key up key down
  , InputDetails value InputValue.KEY_UP
  ("keyUp") InputValue.KEY_DOWN ("keyDown") "KeyDown" .
- x NavigationCode.UP ("up"), NavigationCode.DOWN ("down"), NavigationCode.LEFT
  ("left"), or NavigationCode.RIGHT ("right"),
  navEquivalent (navigation equivalent) .
  NavigationCode(scaleform.clik.ui.NavigationCode)

FocusHandler InputDelegate InputEvent , focus
InputEvent
.
handleInput() .

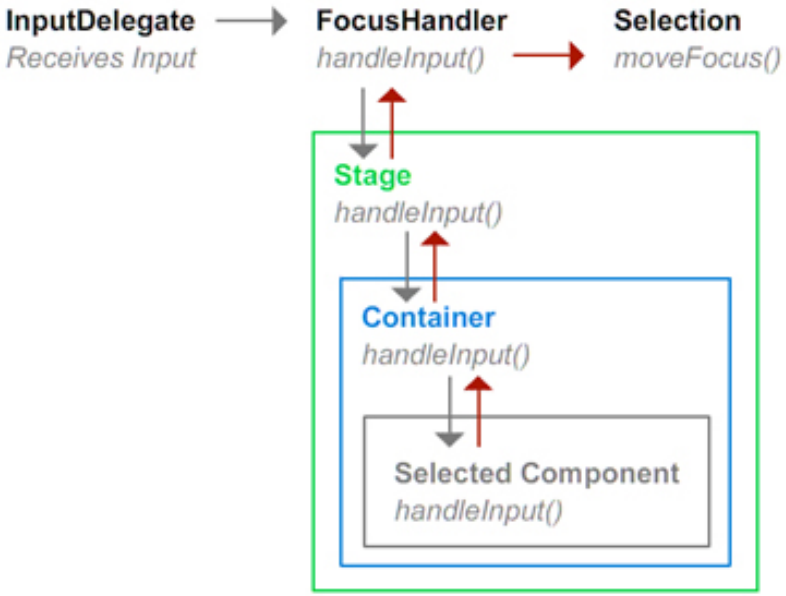


그림 59: handleInput() 체인

InputEvent true .

event.isDefaultPrevented() .

```
override public function handleInput(event:InputEvent):void {
        // Check whether the event has already been handled by another component.
        if (event.isDefaultPrevented()) { return; }
        var details:InputDetails = event.details;

        switch (details.navEquivalent) {
            case NavigationCode.ENTER:
                if (details.value == InputValue.KEY_DOWN) {
                    // Do something.
                    event.handled = true;
                }
                    break;
            default:
                    break;
        }
}
```

InputEvent .bubble ActionScript 3

. ScrollingList ,

InputEvent . Enter

ListItemRenderer ButtonEvent.CLICK

.

```
override public function handleInput(event:InputEvent):void {
    if (event.isDefaultPrevented()) { return; }
    // Pass on to selected renderer first
        var renderer:IListItemRenderer = getRendererAt(_selectedIndex,
_scrollPosition);
    if (renderer != null) {
            // Since we are just passing on the event, it won't bubble, and should
properly stopPropagation.
            renderer.handleInput(event);
        if (event.handled) { return; }
    }
        . . .
}
```

InputDelegate.instance

. FocusHandler ,

.

```
InputDelegate.instance.addEventListener( InputEvent.INPUT, handleInput);
function handleInput(e:InputEvent):void {
    var details:InputDetails = e . details;
    if (details.value = Key. TAB) { doSomething(); }
}
```

InputDelegate.instance

. FocusHandler ,

.

4.6.2 다중 마우스 커서

(Wii™) . CLIK

SWF .

, .

controllerIdx , mouseIndex

.

```
myButton. addEventListener( ButtonEvent.CLICK, onCursorClick);
function onCursorClick(event:ButtonEvent):v oid {
```

```
            trace(event.controllerIdx);
        }
```

4.7

.

.

.

,

.

그림 60: CLIK 컴포넌트는 특정 속성이 변경되면 자동적으로 무효화한다.

4.7.1 사용예와 가장 좋은 연습(Usage and Best Practices)

set), UIComponent.invalidate()

, UIComponent.draw() .

, draw() (: , ,)

isInvalid()

. invalidate() draw()

. isInvalid()

draw() , invalidate() invalidateData(),

invalidateSize() invalidate() .

invalidate() draw() .

. invalidate()

(Event.RENDER)

(Event.ENTER_FRAME) draw() draw()

.

draw() .

UIComponent.validateNow() .

4.8

CLIK 2 .

1. reflowing

. .

2. (counterscaling) .

.

, scale9grid, .

Flash scale9Grid reflow . Flash/Flex

"skin" . Reflowing

container .

Scaleform scale9Grid CLIK

. layered

. CLIK

, ,

. . setups(

...) container . ,

Reflow .

IDE , setSize() , width, height

. IDE .

MovieClips LivePreview .

Scaleform Player

. CLIK

.

4.8.1 Scale9Grid

. Scaleform Player Scale9Grid
MovieClip scale9grid . , MovieClip
. scale9grid
Scaleform .

MovieClip scale9grid
. scale9grid
.

4.8.2 Constraints

Constraints(scaleform.clik.utils.Constraints)
.
, . ,
ScrollBar ,
. Constraints .

ScrollBar initialize () constraints ,
, . draw() constraints
, .
draw() , (
).

```
override protected function initialize():void {
        super.initialize();

    // The upArrow doesn't need a constraints, since it sticks to top left.
    if (downArrow) {
        constraints.addElement("downArrow", downArrow, Constraints.BOTTOM);
    }
    constraints.addElement("track", track, Constraints.TOP |
                                Constraints.BOTTOM);
}

override protected function              preInitialize():void {
```

```
    constraints = new Constraints(this, ConstrainMode.REFLOW);
}

protected function draw():void {
    constraints.update(_width, _height);
}
```

4.9 (Components and Data Sets)

dataProvider . dataProvider

, IDataProvider(scaleform.clik.interfaces.IDataProvider) API

.

dataProvider .

dataProvider .

, .

dataProvider CoreList (ScrollingList, TileList), OptionStepper,

DropdownMenu .

(

x *requestItemRange:* dataProvider .
ScrollingList .
x *indexOf:* .
x *invalidate:* dataProvider
.
"dataChange" . dataProvider length
.

dataProvider .
2
.
dataProvider ExternalInterface.call .

4.10

Scaleform CLIK Tween Scaleform
Tween (scaleform.clik.motion.Tween) . CLIK Tween
fl.transitions.easing.* .

tween Tween() (1000 1), Sprite, tween
, tween Tween
. / .
- *ease*: . .*easeParam*: .
.
- *onComplete*: . Tween .
- *onChange*: . Tween (
tick/frame).
- *data*: . Tween (Tween
).*nextTween*: Tween . Tween
. Tween Tween pause=false
.
- *frameBased*: Boolean . .
- *delay*: . 1000 1 tween .

- *fastTransform*: Boolean .
  .
- *paused*: Boolean . Tween .
  tween MovieClip
  tweening . tween
  .

4.10.1 사용예와 가장 좋은 연습(Usage and Best Practices)

Tween MovieClip MovieClip
tween .

tween , Tween
onComplete . Tween
.

```
import fl.transitions.easing.*;
import    scaleform.clik.motion.Tween;

// Perform a 1 second tween from the current horizontal position and
// alpha values to the ones specified
var   myTween:Tween   = new Tween(300,
                          myMovieClip,
                          { x:100, y:100, alpha:0 },
                      { ease:Strong.easeOut,
                      onComplete:onCompleteCallback} );

function      onCompleteCallback(t:Tween):void          { // Do something. }
```

4.11

CLIK
. , CLIK
UI CLIK
.

: Scaleform v4.0.14 CLIK AS3 API

. CLIK AS3 , , ,

Scaleform (Scaleform

. ,

, .).

CLIK .

- **scaleform.clik.layout.Layout**: .layoutData Sprite .
- **scaleform.clik.layout.LayoutData**: Sprite

.

.

CLIK .

.

1. Sprite/MovieClip DisplayObjectContainer .
2. LayoutData Sprite/MovieClip .layoutData

.

3. Sprite/MovieClip LayoutData .
4. . 1 þ

Layout_FullScreen_Demo

.

4.11.1 레이아웃

/

. Sprite(Sprite)

.

. (myWindow.myMovieClip

myWindow.layoutInstance) myMovieClip

layoutInstance 05790571#

layoutData.layoutIndex Sprite

. layoutIndex .

Flash

. (

0), .

.rect

.

Sprite/MovieClip LayoutData

.

4.11.1.1

rect	" " . x, y, . width != 0 ( ), MovieClip x, y, . width == 0 (addChild()), x, y, . .rect .
tiedToStageSize	true , false .
tiedToParent	true , false .
hidden	true , false . visible = false .

4.11.2 LayoutData

. scaleform.clik.layout.Layout

.

Sprite . :
LayoutMode.ALIGN_NONE, LayoutMode.ALIGN_LEFT,
LayoutMode.ALIGN_RIGHT.

Sprite . :
TextFieldAutoSize.NONE LayoutMode.ALIGN_NONE,
LayoutMode.TOP, LayoutMode.BOTTOM

.

relati ToH Spr B .

m JPp;,

	(Layout.tiedToStageSize == true). 4:3 "LayoutData.offsetHashV[LayoutData.ASPECT_RATIO_4_3] = 70;" Sprite 70px .
relativeToH	Sprite Sprite . null Sprite .
relativeToV	Sprite Sprite . null Sprite .
layoutIndex	Sprite Sprite . Sprite Sprite relativeToH relativeToV . layoutIndex (-1), .
layoutIdentifier	DisplayObjectContainer LayoutData . (null), . .

4.12

CLIK PopUpManager (scaleform.clik.managers.PopUpManager)

.

4.12.1 사용예와 가장 좋은 연습(Usage and Best Practices)

PopUpManager .

ID . initProperties

.

x y . relativeTo Sprite

Sprite .

```
import scaleform.clik.managers.PopUpManager;
PopUpManager. createPopUp( " PopupLinkageID" , {alpha:.5}, 100, 100
myMovieClip);
```

Scaleform

5 사용예

CLIK .

5.1

CLIK

.

5.1.1 2 개의 textField 를 가진 ListItem 렌더러

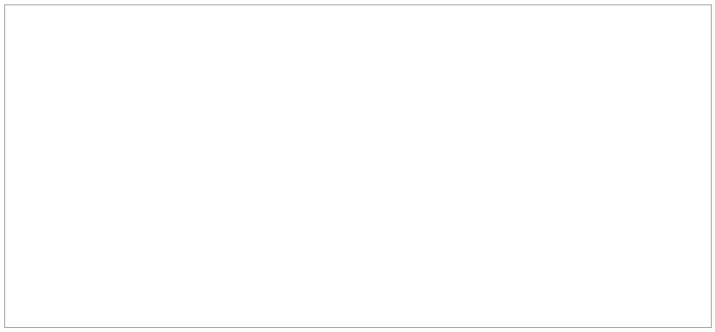


그림 65: 2 개의 레이블을 포함한 리스트 아이템을 보여주는 ScrollingList

ScrollingList (row) ListItemRenderer .
ListItemRenderer textField . textField
. textField .
2 textField .

. 2 textField ListItemRenderer
. ListItemRenderer ScrollingList .
2 textField .
dataProvider .

```
list.dataProvider = [{fname: "Michael", lname: "Jordan"},
                     {fname: "Roger",   lname: "Federer"},
                     {fname: "Michael", lname: "Schumacher"},
                     {fname: "Tiger",   lname: "Woods"},
                     {fname: "Babe",    lname: "Ruth"},
                     {fname: "Wayne",   lname: "Gretzky"},
                     {fname: "Usain",   lname: "Bolt"}];
```

2 (fname lname) . 2
textField 2 .

ListItemRenderer textField , 2 textField
. ListItemRenderer .
MyItemRenderer , ListItemRenderer 2
textField . (FLA
MyItemRenderer.as)

```
import scaleform.clik.controls.ListItemRenderer;

class MyItemRenderer extends ListItemRenderer {

    public var textField1:TextField;
    public var textField2:TextField;

    public function MyItemRenderer() { super(); }

    override public function setData(data: Object):v oid {
        this. data = data;
        textField1.text = data ? data.fname : "";
        textField2.text = data ? data.lname : "";
    }

    override protected function updateAfterStateChange():v oid {
        super.updateAfterStateChange();
        textField1.text = data ? data.fname : "";
        textField2.text = data ? data.lname : "";
    }
}
```

ListItemRenderer setData updateAfterStateChange
(override) . setData
(ScrollingList, TileList,) . ListItemRenderer
textField . , MyItemRenderer textField ,
. updateAfterStateChange

, ListItemRenderer .
textField .

dataProvider
ListItemRenderer . , ListItemRenderer
. ListItemRenderer 'textField1'
'textField2' 2 textField . , ID
MyItemRenderer . MyItemRenderer
itemRenderer 'ListItemRenderer' 'MyItemRenderer' .

FLA . 2 . fname
lname dataProvider .

5.1.2 픽셀별 스크롤 뷰

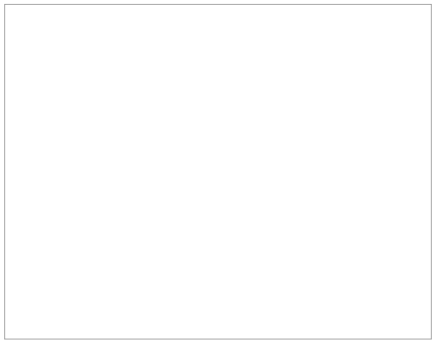


그림 66: CLIK 요소를 컨텐츠로 포함하고 있는 픽셀별 스크롤 뷰

. CLIK
ScrollBar .

, .
.
.
.

- x *actions*: .
- x *scrollbar*: CLIK ScrollBar . 'sb' .

x *mask*: MovieClip . (mask)
.

x *content*: .

x *background*:

, .
. 'content' ,
y (y-value) 0 . .
.

scrollview
. *code* .

```
// 133 is the view size (height of the mask)
sb.setScrollProperties(1, 0, content.height - 133);
sb.position = 0;

sb.addEventListener( Event.SCROLL, onScroll, false, 0, true);
function onScroll(e:Event) {
    content.y = - sb.position;
}
```

Scrollbar .
setScrollProperties
, . scrollbar . .
Scrollview scrollbar .

6 자주 묻는 질문

1. Flash Studio 에서 게시할 때 다음과 같은 오류 메시지를 수신했습니다.

Quote:

> 1152: A conflict exists with inherited definition scaleform.clik... in namespace public.

. ActionScript .FLA

"Automatically Declare Stage Instances"

.FLA

ActionScript Stage

( ) CLIK CLIK

.

File -> Publish Settings -> "Flash" tab -> Actionscript 3.0 Settings .FLA

"Automatically Declare Stage Instances" “Automatically declare stage

instances” . Flash Studio .FLA

.

2. 다른 .FLA 라이브러리에서 CLIK 구성 요소를 가져오는 중입니다. 스테이지에 구성 요소의 인스턴스를 두고 조사 가능한 프로퍼티를 변경하자 다음과 같은 오류 메시지를 수신했습니다.

Code:

> 1046: Type was not found or was not a compile-time constant: ...

.

.

x .

x ActionScript .
x ActionScript . ,
. ActionScript . ,
MyButtonSymbol myButton
myButton:MyButtonSymbol myButton:Button (scaleform.clik.controls.) .

3. 라이브러리에 같은 기본 클래스에서 얻은 두 개의 구성 요소가 있습니다. .SWF 파일을 게시할 때 다음과 같은 오류를 수신했습니다.

Symbol 'MySymbol', Layer 'actions', Frame 10, Line 1 -- 1024: Overriding a function that is not marked for override.

Symbol 'MySymbol', Layer 'actions', Frame 20, Line 1 -- 1024: Overriding a function that is not marked for override.

Symbol 'MySymbol', Layer 'actions', Frame 30, Line 1 -- 1024: Overriding a function that is not marked for override.

.

"Button" "scaleform.clik.controls.Button" .
Symbol .

.

.

.

7 잠재적인 위험

1.

.

.

.

, getDefinitionByName()

.